

Python Complete Practice Guide

Data Analyst Roadmap: Days 1-8

■ Day	■ Topics	■ Questions
Day 1	Variables, Data Types, Operators	20
Day 2	If-Else, Comparison, Logic	20
Day 3	For Loop, While Loop, Range	20
Day 4	Functions, Arguments, Return	20
Day 5	Lists, Tuples, Indexing	25
Day 6	Dictionary, Sets, Strings	20
Day 7	File Handling, Exceptions	15
Day 8	Mixed Revision (All Topics)	30
TOTAL		170+ Question

■ **How to Use This Guide:** Practice one day at a time. Solve without looking at solutions. Debug and understand errors.

■■ **Recommended Time:** 20-30 minutes daily for each day's questions.

DAY 1: Variables, Data Types, Input(), Print(), Operators (20 Questions)

Q1: Create three variables (name, age, city) and print them as: 'My name is [name], I am [age] years old, and I live in [city]'.

Q2: Create variables with different data types (int, float, string, bool) and use type() to identify each.

Q3: Write a program that takes two numbers as input and displays their sum, difference, product, quotient, and remainder.

Q4: Convert Celsius to Fahrenheit using $F = (C \times 9/5) + 32$. Take temperature in Celsius as input.

Q5: Evaluate: `print(2 + 3 * 4 - 5 / 2)`. Explain the order of operations.

Q6: Take two strings as input, concatenate them, and display the length and reversed version.

Q7: Swap two variables without using a third variable.

Q8: Create a BMI calculator: $BMI = \text{weight} / (\text{height}^2)$. Classify as underweight/normal/overweight.

Q9: Find the area and perimeter of a rectangle given length and width.

Q10: Calculate simple interest: $SI = (P \times R \times T) / 100$.

Q11: Convert kilometers to miles and vice versa.

Q12: Calculate the total cost with a 10% discount given the original price.

Q13: Find the average of three numbers.

Q14: Calculate the volume and surface area of a sphere given radius.

Q15: Create a simple age calculator from birth year.

Q16: Calculate electricity bill: $\text{units} \times \text{rate} + \text{fixed charge}$.

Q17: Currency converter: convert INR to USD, EUR, GBP.

Q18: Calculate percentage increase/decrease between two values.

Q19: Find the compound interest using $CI = P(1 + R/100)^T$.

Q20: Calculate the nth term of an arithmetic sequence: $a_n = a + (n-1)d$.

DAY 2: If-Else, Comparison, Logical Operators (20 Questions)

Q1: Check if a number is positive, negative, or zero.

Q2: Check eligibility to vote (age ≥ 18).

Q3: Find the largest of three numbers using if-elif-else.

Q4: Assign grades based on marks: A (90-100), B (80-89), C (70-79), D (60-69), F (<60).

Q5: Check if a year is a leap year.

Q6: Check if a number is between 1 and 100 (inclusive).

Q7: Check if a number is even/odd AND positive/negative.

Q8: Write a simple calculator (+, -, *, /, %).

Q9: Check if a person can get a loan (age ≥ 21 AND income ≥ 100000).

Q10: Determine if a triangle is valid given three sides.

Q11: Check if a number is divisible by both 3 and 5.

Q12: Determine if a person is eligible for senior citizen discount (age ≥ 60).

Q13: Check password strength (length ≥ 8 AND has uppercase AND has numbers).

Q14: Find the maximum temperature of the day (given morning, afternoon, evening temps).

Q15: Check if a number is a perfect square.

Q16: Determine overtime payment: if hours > 8 , extra 1.5x pay.

Q17: Check if three numbers can form a right triangle.

Q18: Determine if a date is valid (considering days in month).

Q19: Check if a person qualifies for a scholarship (score ≥ 85 AND income ≤ 500000).

Q20: Assign T-shirt size: XS (< 35), S (35-40), M (40-50), L (50-60), XL (> 60).

DAY 3: For Loop, While Loop, Range (20 Questions)

Q1: Print numbers from 1 to 10 using for loop with range().

Q2: Print multiplication table of a number (1-10).

Q3: Calculate sum of first N natural numbers using while loop.

Q4: Calculate factorial of a number using a loop.

Q5: Print a pyramid pattern: *, **, ***, ****, *****

Q6: Print all even numbers between 1 and 50.

Q7: Generate first 10 Fibonacci numbers.

Q8: Print numbers 1-20, skip multiples of 3 (continue), stop at 15 (break).

Q9: Print reverse series from 50 to 1.

Q10: Calculate sum of even numbers from 1 to 100.

Q11: Print times table of 5 (1x5 to 10x5).

Q12: Find sum of digits of a number.

Q13: Print numbers that are divisible by both 3 and 5 up to 100.

Q14: Print a hollow square pattern of size 5x5.

Q15: Calculate power of a number using loop.

Q16: Print all perfect squares up to 100.

Q17: Count and print vowels in a string using loop.

Q18: Print ASCII values of characters a-z.

Q19: Find the largest number in a range 1-1000 divisible by 7.

Q20: Generate and print prime numbers up to 50.

DAY 4: Functions, Arguments, Return (20 Questions)

Q1: Write a function that returns the square of a number.

Q2: Write a function to find the maximum of three numbers.

Q3: Create a function to convert temperature (C to F and F to C).

Q4: Write a function to reverse a string.

Q5: Write a function to check if a number is prime.

Q6: Write a function with default parameters to calculate area (length, width=length).

Q7: Write a function that returns sum, difference, and product of two numbers.

Q8: Write a recursive function to calculate factorial.

Q9: Write a function to find the GCD of two numbers.

Q10: Write a function to count vowels in a string.

Q11: Write a function to check if a string is a palindrome.

Q12: Write a function to find the LCM of two numbers.

Q13: Write a function to convert decimal to binary.

Q14: Write a function to check if a number is Armstrong number.

Q15: Write a function to sum all elements in a list.

Q16: Write a function to find average of a list of numbers.

Q17: Write a function to remove duplicates from a list.

Q18: Write a function to check if a list is sorted.

Q19: Write a function to rotate a list by k positions.

Q20: Write a function to find the second largest number in a list.

DAY 5: Lists, Tuples, Indexing (25 Questions)

- Q1:** Create a list of 5 numbers. Access 1st, 3rd, and last elements.
- Q2:** Given [1,2,3,4,5,6,7,8,9,10], slice: first 5, last 3, every alternate.
- Q3:** Create list [3,1,4,1,5]. Use append(), insert(), remove(), sort(), reverse().
- Q4:** Write a function to calculate sum and average of list elements.
- Q5:** Write a function to find index of an element in a list.
- Q6:** Create a tuple (5,10,15) and list [5,10,15]. Try to modify both.
- Q7:** Find and count duplicate elements in a list.
- Q8:** Find highest, lowest, and average marks from a list. Count students with marks > 80.
- Q9:** Merge two lists without using + operator.
- Q10:** Find common elements between two lists.
- Q11:** Remove duplicates from a list while preserving order.
- Q12:** Flatten a nested list [[1,2], [3,4], [5,6]].
- Q13:** Split a list into n equal parts.
- Q14:** Find elements that appear only once in a list.
- Q15:** Rotate a list by k positions.
- Q16:** Find the majority element in a list.
- Q17:** Pair elements from two lists: [1,2,3] + [a,b,c] = [(1,a), (2,b), (3,c)].
- Q18:** Find subarray with maximum sum (Kadane's algorithm).
- Q19:** Check if list is sorted in ascending and descending order.
- Q20:** Move all zeros to the end of a list.
- Q21:** Find kth largest element in a list.
- Q22:** Check if two lists have same elements (regardless of order).
- Q23:** Find missing number in a list of 1 to n.
- Q24:** Reverse list without using reverse() or slicing.
- Q25:** Find intersection and union of two lists.

DAY 6: Dictionary, Sets, Strings (20 Questions)

- Q1:** Create a dictionary with student info (name, age, grade, marks). Access and modify values.
- Q2:** Demonstrate dict methods: keys(), values(), items(), get(), pop().
- Q3:** Count frequency of each character in a string using a dictionary.
- Q4:** Create two sets and perform union, intersection, difference operations.
- Q5:** Demonstrate string operations: upper(), lower(), capitalize(), split(), join(), replace().
- Q6:** Use a set to find unique elements in a list.
- Q7:** Create a contact book (dictionary). Add, search, delete contacts.
- Q8:** Check if a string is a palindrome (ignore spaces and case).
- Q9:** Count word frequency in a sentence using a dictionary.
- Q10:** Merge two dictionaries.
- Q11:** Find common elements between two sets.
- Q12:** Sort a dictionary by values.
- Q13:** Create nested dictionary for student grades: {name: {subject: grade}}.
- Q14:** Check if two strings are anagrams.
- Q15:** Find longest word in a string.
- Q16:** Check if a string contains only alphanumeric characters.
- Q17:** Convert a list of tuples to a dictionary.
- Q18:** Find uncommon elements between two sets.
- Q19:** Check if a string is a substring of another.
- Q20:** Create frequency counter for elements in a list using dictionary.

DAY 7: File Handling, Exceptions (15 Questions)

- Q1:** Write a program to create a file and write 5 lines of text.
- Q2:** Read a file line by line and print each with line number.
- Q3:** Divide two numbers with try-except to handle ZeroDivisionError.
- Q4:** Handle FileNotFoundError, ValueError, TypeError in single try-except block.
- Q5:** Read data from file, count lines/words/characters. Write result to new file.
- Q6:** Write employee records (name, ID, salary) to file. Read and display formatted.
- Q7:** Write try-except-finally block: open file, read, handle errors, close properly.
- Q8:** Validate user input (integer, age > 0) using try-except. Store in file.
- Q9:** Read CSV file and print data in formatted table.
- Q10:** Copy contents from one file to another with error handling.
- Q11:** Read file and replace a word with another. Write to new file.
- Q12:** Implement custom exception for age validation.
- Q13:** Read numbers from file and calculate their sum/average/max/min.
- Q14:** Append data to existing file with try-except.
- Q15:** Create a program that reads file and counts specific word occurrences.

DAY 8: REVISION - Mixed Python Concepts (30 Questions)

- Q1:** Create ATM menu with Check Balance, Withdraw, Deposit, Exit options.
- Q2:** Create student management system. Add, delete, search, display students.
- Q3:** Classify 20 numbers into even/odd, positive/negative. Display statistics.
- Q4:** Read marks from file. Calculate average, highest, lowest, grade distribution.
- Q5:** Create registration form with validation: email, age (18-65), phone format.
- Q6:** Create inventory dictionary. Add, remove, update stock with exceptions.
- Q7:** Print pyramid/diamond pattern based on user input.
- Q8:** Build calculator with history. Support +, -, *, /, %. Save to file.
- Q9:** Merge two lists, find common, remove duplicates, sort, find missing elements.
- Q10:** Read text file. Count words, characters, lines. Find most frequent words.
- Q11:** Create a to-do list application: add, remove, mark complete, save to file.
- Q12:** Implement number guessing game with hints and score tracking.
- Q13:** Create a bank account system with deposit, withdrawal, interest calculation.
- Q14:** Implement rock-paper-scissors game against computer.
- Q15:** Create password manager that stores encrypted passwords in file.
- Q16:** Build a simple library system with add book, search, borrow, return.
- Q17:** Create grade calculator for multiple subjects with weighted averages.
- Q18:** Implement hangman word guessing game.
- Q19:** Create employee salary calculator with tax and deductions.
- Q20:** Build a quiz application that reads questions from file and tracks score.
- Q21:** Create hotel room booking system with pricing and availability.
- Q22:** Implement a traffic light system simulator.
- Q23:** Create contact management system with add, search, update, delete.
- Q24:** Build a simple weather application storing and analyzing temperature data.
- Q25:** Create a sudoku solver using backtracking algorithm.
- Q26:** Implement binary search and linear search comparison.
- Q27:** Create bubble sort and insertion sort implementations with step tracking.
- Q28:** Build a file encryption/decryption tool.
- Q29:** Create a phone directory with call logs and statistics.
- Q30:** Implement a simple web scraper with error handling and data storage.

■ Practice Tips & Guidelines

- ✓ **Daily Practice:** Dedicate 20-30 minutes to solve questions for each day.
- ✓ **No Copy-Paste:** Write code from scratch, don't copy from solutions.
- ✓ **Understand Errors:** When you get an error, debug and understand why.
- ✓ **Test Edge Cases:** Test with various inputs including boundary values.
- ✓ **Code Review:** After solving, review your code for improvements.
- ✓ **Track Progress:** Mark completed questions and track time taken.
- ✓ **Upload to GitHub:** Create GitHub repository with your solutions.
- ✓ **Revision:** Day 8 is for revision - revisit difficult topics.
- ✓ **Time Management:** Allocate: Day 1-7 (20-30 min each), Day 8 (30-45 min).
- ✓ **Documentation:** Add comments in your code explaining the logic.

■ Progress Summary:

Day	Questions	Topics	Difficulty
1	20	Variables, Data Types	Basic
2	20	Conditionals	Basic
3	20	Loops	Basic → Intermediate
4	20	Functions	Intermediate
5	25	Lists & Tuples	Intermediate
6	20	Dict, Sets, Strings	Intermediate
7	15	File & Exceptions	Intermediate
8	30	Full Project-Based	Intermediate → Advanced

Next Steps: After completing this Python practice (Days 1-8), you'll be ready for Days 9-15 (NumPy & Pandas).